Design and Fabrication of Low-Temperature Continuous-Flow Cryostat System

Nguyen Xuan Tai, Nguyen Tien Dat, Tran Phan Sang, Nguyen Minh Tam

Department of Advanced Materials Science and Nanotechnology, University of Science and Technology of Hanoi (USTH)
Instructors: Dr. Nguyen Van Quyen (AMSN, USTH) · Nguyen Van Huan (An Tai Loc Co., Ltd.)

1. Motivation & Physical Principles

Cryogenic environments (77 K – 300 K) are indispensable for investigating low-dimensional quantum transport, semiconductor devices, and phase transitions. Conventional commercial cryostats are cost-prohibitive for flexible laboratory prototyping.

Core Objective: Develop a customizable, modular continuous-flow cryostat operating between 77 K and 300 K with high-vacuum isolation ($P < 10^{-4}$ mbar) and in-situ 4-probe electrical transport tracking.

Thermodynamic Heat Balance:

$$\dot{Q}_{\text{total}} = \dot{Q}_{\text{cond}} + \dot{Q}_{\text{rad}} + \dot{Q}_{\text{conv}}$$

- **Conduction Leakage:** Fourier's Law with temperature-dependent $k(T)$:

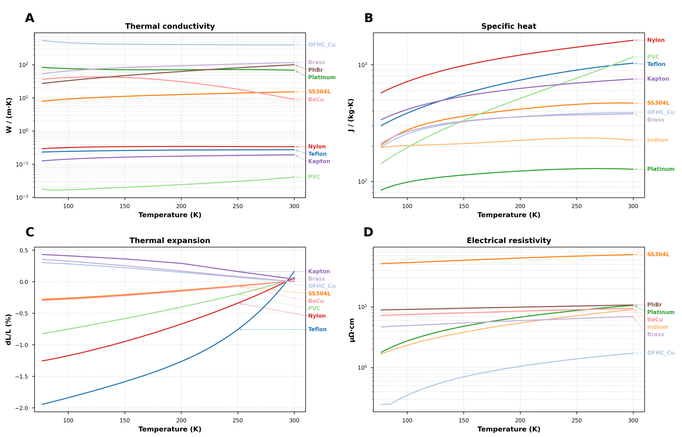
$$\dot{Q}_{\text{cond}} = \frac{A}{L}\int_{T_C}^{T_H} k(T)\,dT$$

- **Radiation Transfer:** Stefan-Boltzmann law with radiation baffles:

$$\dot{Q}_{\text{rad}} = \epsilon\sigma A(T_H^4 - T_C^4)$$

- **High Vacuum Molecular Flow:** At $P \leq 10^{-4}$ mbar, molecular mean free path exceeds chamber dimensions $\implies \dot{Q}_{\text{conv}} \approx 0$.

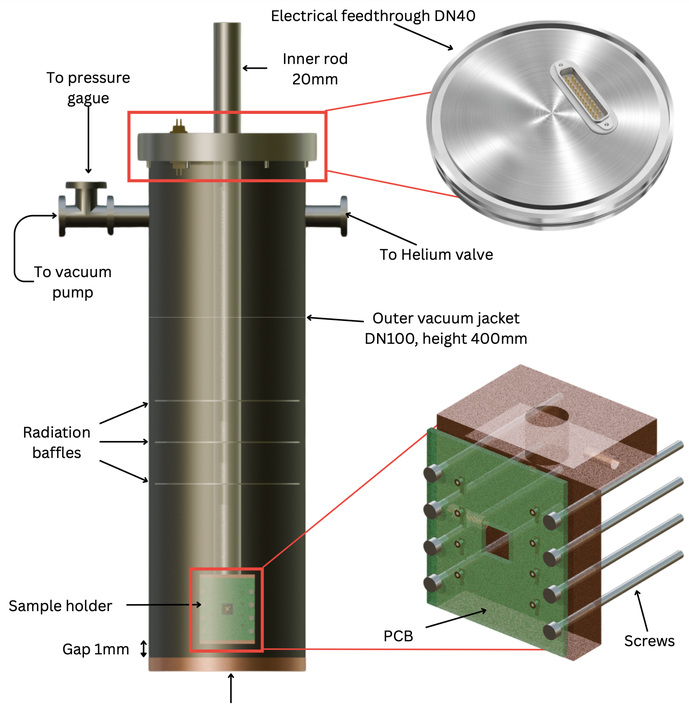
2. Materials & Thermal Coupling



Thermal conductivity $k(T)$ and electrical resistance across materials.

- **OFHC Copper (C10100):** Cold finger and sample stage ($k \approx 390$ W/(m · K)

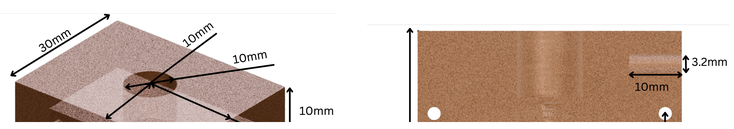
3. Mechanical Architecture & CAD



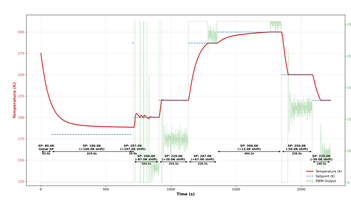
Full 3D cross-sectional CAD assembly of the cryostat.

- **Modular Vacuum Enclosure:** SS304L body with standard KF-40 flanges, viton O-rings, and quick-release centering clamps.
- **Cryogenic Delivery Tube:** Continuous liquid nitrogen delivery tube entering the cold head with minimal parasitic conductive bridges.

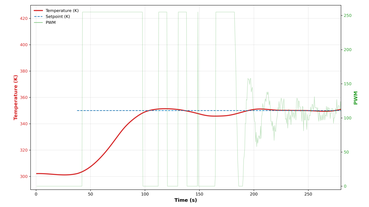
4. Sample Stage & 4-Probe Interface



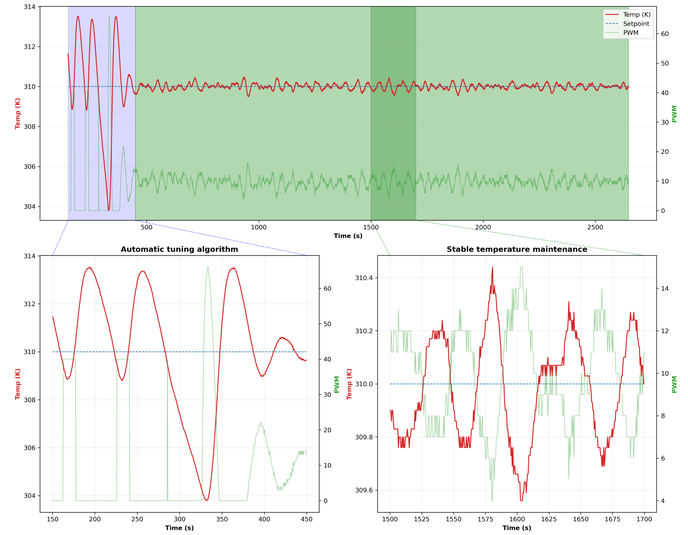
5. Experimental Results & Thermal PID



Continuous-flow cooldown



Vacuum pump-down curve



PID thermal regulation (±0.4 K) across multi-step setpoints.

- **Dynamic Thermal Tracking:** Åström-Hägglund relay auto-tuning with Tyreus-Luyben rules smoothly managed thermal inertia.
- **Thermal Stability:** Maintained setpoints with ±0.4 K steady-state fluctuation from 160 K to 300 K without sustained oscillation.

6. System Specifications & Conclusion

Parameter	Demonstrated Value
Temperature Range	160 K – 300 K (Tested) / 77 K (Baseline LN2)